on Earth in late 2015, and the announcement, given in early 2016, has once again left the world speechless. Once more, the seemingly mad predictions of Einstein's theory turn out to be precisely true.

And further still, the theory predicts that the universe is expanding and emerged from a cosmic explosion 14 billion years ago – a subject I will discuss in more detail shortly.

This rich and complex range of phenomena – bending of rays of light, modification of Newton's force, slowing down of clocks, black holes, gravitational waves, expansion of the universe, the Big Bang – follow from understanding that space is not a dull, fixed container but possesses its own dynamic, its own 'physics', just like the matter and the other fields it contains. Democritus himself would have smiled with pleasure, had he been able to see that his idea of space would turn out to have such an impressive future. It is true that he termed it non-being, but what he meant by being (δέν) was matter; and he wrote that his non-being, the void, nevertheless 'has a certain physics (φύσιν) and a substantiality of its own'.[fn14] How right he was.

Without the notion of fields introduced by Faraday, without the spectacular power of mathematics, without the geometry of Gauss and Riemann, this 'certain physics' would have remained incomprehensible. Empowered by new conceptual tools and by mathematics, Einstein writes the equations which describe Democritus's void and finds for its 'certain physics' a colourful and amazing world where universes explode, space collapses into bottomless holes, time slows down in the vicinity of a planet, and the boundless expanses of interstellar space ripple and sway like the surface of the sea …

All of this sounds like a tale told by an idiot, full of sound and fury, signifying nothing. And yet, instead, it is a glance towards reality. Or better, a glimpse of reality, a little less veiled than our blurred and banal everyday view of it. A reality which seems to be made of the same stuff our dreams are made of, but which is nevertheless more real than our clouded daily dreaming.

And all this is the result only of an elementary intuition – that spacetime and the gravitational field are one and the same thing – and a simple equation which I can't resist copying out here, even if most of my readers will certainly not be able to decipher it. I do so, anyway, in the hope that they might be able to catch a glimpse of its beautiful simplicity:

$$R_{ab} - \frac{1}{2} R g_{ab} + \Lambda g_{ab} = 8\pi G\, T_{ab}$$